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Unit 3 - Designing Solutions

PDF Documents

PDF Document 1: 1738070331429-Chapter 7 - Unit 3 - Designing Solutions.pdf

This PDF document is included in the unit materials.

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UNIT 3 DESIGNING SOLUTIONS

● Programming Concepts

RANDOM

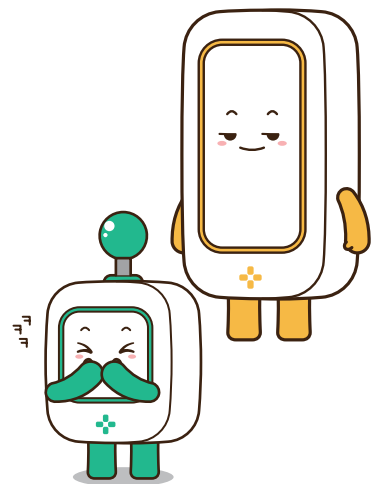
The Random block is used to generate a single random number within a specified range. This feature is particularly useful when you want to introduce variability or unpredictability into your projects.



You can assign the output from a Random block to a variable, creating what's known as a random variable. Alternatively, you can use the random number directly within conditional statements to influence program behavior dynamically.

TIPS

In this example, the value for the red color in an LED light is chosen randomly by using the random block. This introduces an element of unpredictability and variety, making the LED's color change dynamically.





MODULO OPERATOR

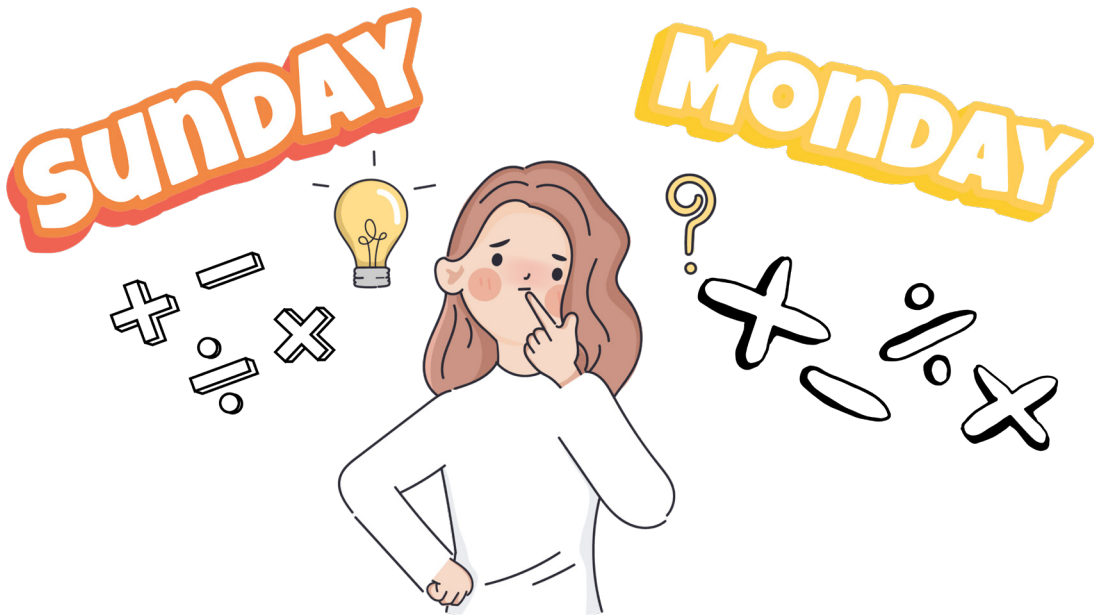
UNDERSTANDING THE MODULO OPERATOR

9 mod 5

The modulo operator is used to determine the remainder of a division operation between two numbers. For example, if you divide 9 by 5, the quotient (the number of times one number is fully contained in the other) is 1, and the remainder is 4. Thus, when using the modulo operator, $9 \bmod 5$ equals 4, because 4 is what remains after dividing 9 by 5 evenly once.

5 mod 5

Similarly, $5 \bmod 5$ equals 0 because 5 divides into 5 exactly once with nothing left over.

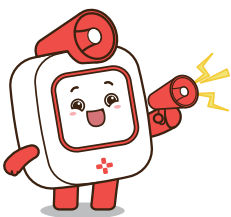


PRACTICAL APPLICATIONS

This operator is incredibly useful in programming, especially for creating repetitive patterns or managing cyclical calculations. For instance, in tracking days of the week, we can represent the days as numbers from 0 to 6 (where 0 could be Sunday, 1 Monday, etc.). If you want to find out what day it will be 100 hours from now, you can calculate $(\text{current_day} + 100) \bmod 7$. This calculation ensures that if the sum exceeds 6 (Saturday), it wraps around to Sunday and continues through the week accordingly.

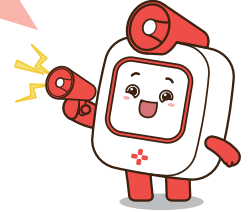
THINK ABOUT IT!

If you compute $'x' \bmod 'y'$, then the possible values for the modulo result range from 0 to $'y' - 1$. This is because the modulo operation gives you the remainder of the division of $'x'$ by $'y'$, and the largest remainder you can have is one less than the divisor ($'y'$).



THINK ABOUT IT!

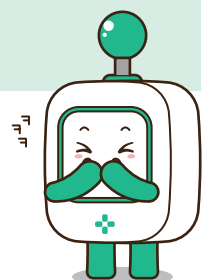
Surprise! Let's dive into some discussion questions with your peers! Since concepts like randomness and the modulo operator can be confusing for those learning them for the first time, let's share insights and ideas with each other!

**Random**

I've heard about randomness in programming, but I'm not sure how it works. Can you explain how randomness is used in programming languages? How can we generate random numbers, and why is it important in coding?

Modulo

I've come across the term 'modulo operator' while learning programming, but I'm not sure what it does. Can you explain how the modulo operator works and why it's useful in programming? Are there any real-world examples where we would use the modulo operator?





Flowcharts & Algorithms

Let's read the flowchart and create the code.

FLOWCHART

